

CompSci 234/NetSys 210
Advanced Topics in Networking
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Unit 8: 360 Video Streaming

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Slide adopted from Prof. Zimmerman's materials

Agenda

- Cameras and Displays
- Projection Models
- Tiled Streaming (PA1)
- Edge Rendering
- Fixation Prediction
- Other Challenges

How to Capture 360 Videos?



Towards Consumer-Graded Cameras

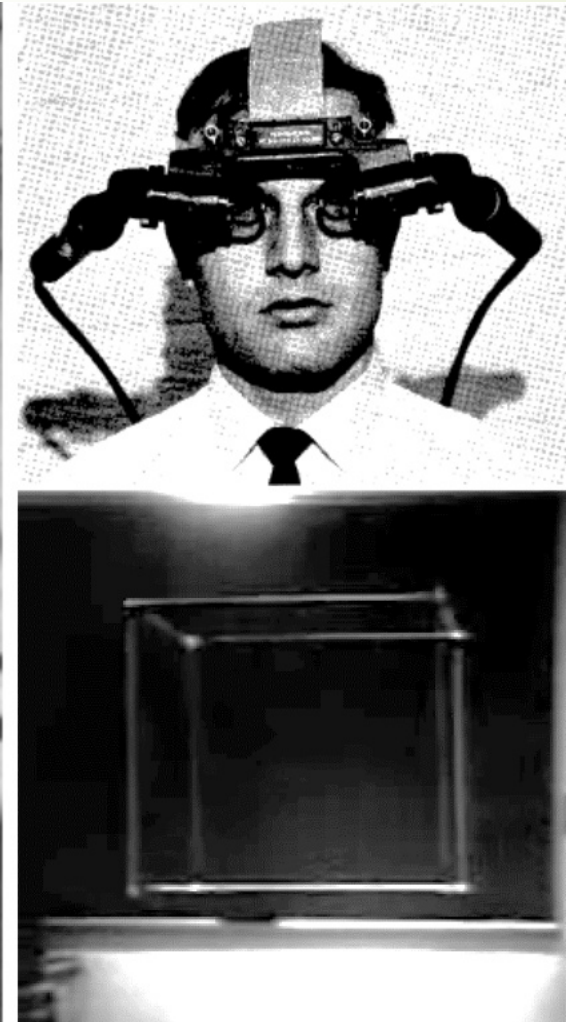
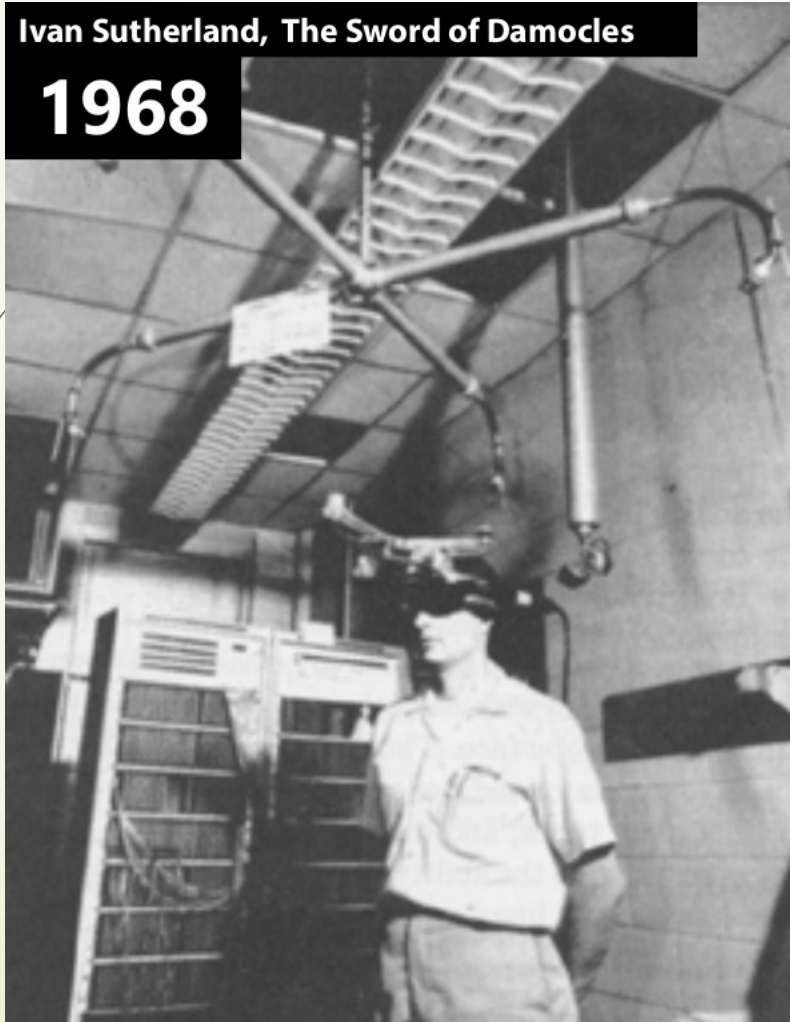
- Live Streaming with 360° cameras
- Examples (2017):
 - Giroptic iO
 - Samsung Gear 360
 - Insta360 Nano/Air



How to Consume 360 Videos?

Ivan Sutherland, The Sword of Damocles

1968



Modern Head-Mounted Displays (HMDs) for Virtual Reality



HTC Vive



Oculus Rift

- Standalone
- Tethered with PC
- Mobile



PSVR